



Exploring user awareness and perceived usefulness of generative AI in higher education: The moderating role of trust

Mai-Lun Chiu¹ 

Received: 30 December 2024 / Accepted: 5 May 2025

© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2025

Abstract

Generative Artificial Intelligence (GAI), with its distinctive capabilities, is inducing profound transformations across multiple industries. In the realm of higher education, centered around human intellect and pedagogical advancement, these emerging technologies are at a pivotal juncture of change. Within the scope of GAI, the dynamics of human-technology interaction are significantly influenced by user cognizance, encompassing an understanding and experiential knowledge of technological features. This study investigates the impact of GAI adoption in higher education, focusing on how user awareness influences perceived utility through the lens of the Status Quo Bias (SQB) theory. The research employs a quantitative approach using Partial Least Squares Structural Equation Modelling (PLS-SEM) to examine the relationships between various factors affecting GAI implementation in educational settings. The findings reveal significant effects on user awareness from two main dimensions: individual commitment to resistance and technology affordance. In the individual commitment to resistance dimension, both sunk cost and privacy concerns demonstrated significant influences on user awareness. Within the technology affordance dimension, explainability, accountability, and transparency showed significant positive effects on user awareness. These results provide valuable insights for higher education institutions implementing GAI technologies, suggesting that successful adoption requires careful consideration of both individual resistance factors and technological affordance features. The findings contribute to the understanding of GAI adoption in educational contexts and offer practical implications for developing effective implementation strategies.

Keywords Generative artificial intelligence (GAI) · Technology affordance · Status Quo Bias (SQB) theory · User awareness

✉ Mai-Lun Chiu
mlchiu@staff.nkuht.edu.tw

¹ General Education Center, National Kaohsiung University of Hospitality and Tourism, No.1, Songhe Rd., Xiaogang District, Kaohsiung 812301, Taiwan

1 Introduction

Artificial Intelligence (AI) has profoundly impacted diverse sectors by streamlining processes, facilitating decision-making, and providing insights based on patterns observed in data (Tekic & Füller, 2023; Yilmaz et al., 2023). While traditional AI models operate on existing data to perform specific tasks, Generative AI (GAI) offers a distinct approach. GAI, particularly when built on architectures such as generative adversarial networks, possesses the unique ability to generate novel outputs by learning from vast datasets (Yilmaz & Yilmaz, 2023; Zastudil et al., 2023). Unlike deterministic AI models, GAI can produce new, previously unseen content, pushing the limits of machine creativity (Niu & Mvondo, 2024; Steele, 2023; Watermeyer et al., 2024).

GAI is profoundly reshaping numerous sectors with its unparalleled capacity to produce novel and meaningful content. In industries such as semiconductors, GAI can be instrumental in optimizing chip designs by analyzing countless permutations and proposing the most efficient configurations that human designers may overlook (Hung, 2023). In supply chain management, GAI offers transformative potential by forecasting demand, optimizing inventory, and even generating innovative product designs based on market trends. This AI-driven evolution promises greater efficiency and reduced operational costs. Nonetheless, vast areas of application remain largely unexplored.

The key to harnessing this potential lies in combining domain-specific knowledge with GAI models, ensuring that AI solutions align with human needs and the intricacies of each sector. GAI's expansion into various domains heralds an era of unprecedented technological innovation. Notably, higher education, traditionally viewed as an epicenter of human intellect and pedagogical development, stands at the cusp of significant transformation driven by these emergent technologies. Yilmaz and Yilmaz (2023) noted that GAI offers a nuanced approach to curriculum design, promising more individualized and optimized learning paths. Kaplan- Rakowski et al. (2023) found that most teachers believe GAI tools benefit students and support teacher development. A moderate correlation exists between teachers' level of GAI integration and the frequency of its use in teaching, suggesting that increased exposure leads to more positive attitudes. While the underlying reasons remain unclear, educators generally appear willing to embrace GAI tools in instruction.

The dynamics of human interactions with GAI technologies are largely shaped by user awareness. This awareness, defined as a cognitive representation of a user's understanding, experience, and knowledge of technological characteristics, plays a critical role in shaping evaluations of a technology's effectiveness. Higher educational environments may intensify this awareness, as both educators and students strive to extract value while remaining wary of potential pitfalls (Kong et al., 2023). The concept of perceived usefulness, as introduced by Davis et al. (1989) in the Technology Acceptance Model (TAM), refers to the extent to which a person believes that using a particular technology will enhance performance (Maheshwari, 2024). When viewed through the lens of user awareness, the relationship becomes particularly intriguing. Heightened awareness can streamline users' cognitive load, enabling them to more effectively recognize the salient features and advantages of a given technology, thereby increasing its perceived usefulness.

However, empirical studies also suggest that a deeper understanding of technological risks may lead users to perceive a technology as less useful (Bilquise et al., 2024; Zhang et al., 2025). In the context of GAI, comprehending its mechanisms, potential, and implications can give users a heightened sense of control and assurance (Fast & Jago, 2020). As users become more familiar with GAI, they tend to rely more on its capabilities while simultaneously expressing concerns about the trustworthiness of algorithms and the risk that their specialized knowledge might be easily replicated by AI systems.

Another important cognitive construct shaping human interaction with GAI is the concept of sunk cost. Petcharat et al. (2023) argued that cognitive mechanisms linked to previous investments, such as time and effort, can influence decision-making. For example, students may cling to traditional learning systems, overlooking potentially more effective GAI-based tools. Moreover, GAI's ability to rapidly assimilate knowledge and the associated privacy concerns raise additional issues. The widespread integration of individual academic contributions with GAI could lead to a homogenization of intellectual output, sparking ethical debates about ownership and the digital dissemination of knowledge.

In the context of technology adoption, attitude closely aligns with perceived usefulness. Studies have shown that users with a deep understanding of a technology's intricacies, along with its potential benefits and drawbacks, may be more inclined to view it skeptically. In the higher education context, understanding GAI's underlying mechanisms can enhance users' sense of mastery and confidence (Memarian & Doleck, 2023). This deeper awareness is likely to reshape their perceptions of GAI's value in academic tasks and research.

Both human cognitive dimensions and technological affordances influence user awareness. To understand the dynamics of this awareness, this study examines how GAI's technical features, specifically *explainability*, *accountability*, and *transparency*, impact user consciousness. *Explainability* refers to the extent to which a model's decision-making processes can be understood by human stakeholders (Shin et al., 2022). *Accountability* entails the responsibility of answering for one's actions, while *transparency* relates to the clarity and accessibility of a system's processes. For instance, if a GAI tool curates personalized study materials, both educators and students are more likely to trust its recommendations if they can understand the underlying logic. Conversely, a lack of transparency may breed skepticism, reducing user confidence and diminishing the tool's perceived utility.

Although previous studies have suggested that higher user awareness can reduce perceived usefulness, they also show that trust in a system increases users' willingness to engage with it (Surahman & Wang, 2022; Hyun & Kim, 2023). Therefore, this study proposes trust as a moderating factor. GAI offers transformative possibilities in higher education, from personalized learning paths to dynamic content generation and assessment tools. The core of the research model is to examine how user awareness, shaped by cognitive mechanisms and technology traits, influences the perceived usefulness of GAI. Trust moderates this relationship, acting as a bridge between awareness and perceived educational utility. The research questions are as follows:

RQ1: How do students' cognitive mechanisms affect their awareness of GAI applications?

RQ2: How does students' understanding of GAI's technological affordances shape their awareness and influence its educational applications?

RQ3: How does trust, as a moderating variable, impact students' awareness of GAI and their perceptions of its usefulness?

2 Theory and literature review background

The Theory of Planned Behavior (TPB) posits that an individual's intention to engage in a behavior is influenced by their attitude, subjective norms (SN), and perceived behavioral control (PBC) (Ajzen, 2002). SN pertains to individual beliefs about the behavior, while PBC mainly involves technology-related beliefs. This study adopts TPB as an overarching theoretical framework to define two key drivers: individual commitment to resistance and technological affordances. This section provides an overview of the literature on cognitive and technological affordances and their association with education. Section 2.1 examines the concept of user awareness in the context of GAI, focusing on how learners understand and interact with GAI systems for educational purposes. Section 2.2 explores the notion of individual commitment to resistance, rooted in status quo bias (SQB) theory, which explains individuals' tendency to prefer existing situations to potential changes, even when alternatives are available. Section 2.3 discusses technology affordance, defined as the potential uses of a technology based on its attributes, emphasizing how it extends learning and perceptual capabilities. It explores how GAI-related affordances, particularly explainability, accountability, and transparency (EAT), facilitate students' information gathering and knowledge creation, supporting constructivist online learning environments. Section 2.4 focuses on perceived usefulness, exploring how learners assess the impact of GAI systems on their learning outcomes and experiences. Section 2.5 highlights the critical role of trust in users' engagement with GAI systems.

3 User awareness

User awareness refers to the state or capability of comprehending the current situation, encompassing an individual's knowledge of their environment, shaped by interactions between the individual and their surroundings (Shin et al., 2022; Taylor & Brisini, 2024). In this study, user awareness refers to the degree of understanding learners have about GAI algorithms and how these can be applied for learning purposes. This understanding evolves dynamically as GAI systems develop and as learners engage more deeply with them (Dalalah & Dalalah, 2023; Duong et al., 2023; Watermeyer et al., 2024). User awareness in the GAI context includes both cognitive recognition of a system's capabilities and the continuous interaction with it, such as exploring GAI features, interpreting its outputs, and adapting its use for effective learning (Bogina et al., 2021; Khosravi et al., 2022; Memarian & Doleck, 2023).

Significantly, awareness also involves recognizing psychological barriers to adopting new technologies. Learners may find it difficult to switch from familiar

systems due to prior investments of time, effort, and established learning strategies (Balakrishnan et al., 2021; Petcharat et al., 2023). User awareness shapes how learners interact with GAI systems, encompassing not only recognition of potential benefits but also an understanding of both the affordances and the risks these technologies may present (Jain & Raghuram, 2024; Akter et al., 2021; Zaroual et al., 2021). These interactions influence learners' behaviors, outcomes, and the broader integration of GAI technologies in educational environments.

4 Individual commitment to resistance

Status Quo Bias (SQB) theory is rooted in psychological theory and applied across decision-making disciplines (Balakrishnan et al., 2021; Wang et al., 2018). It posits that individuals tend to maintain their current situation even when better alternatives are available (Samuelson and Zeckhauser, 1988). A key component of SQB is individual commitment, which refers to the internal motivation individuals sustain when resisting the adoption of novel technologies. Resistance to new technology often arises for two primary reasons. First, individuals may be reluctant to switch due to already invested resources, time, or effort in the existing system, representing a sunk cost. Second, individuals may fear regret when adopting new systems, particularly concerning potential privacy breaches or the misuse of personal data and intellectual property. Prior research has identified both sunk costs and privacy concerns as significant contributors to resistance in adopting new systems (Kulkov et al., 2023; Liu & Tao, 2022; Petcharat et al., 2023).

5 Sunk cost

Sunk costs refer to investments that have already been made and cannot be recovered. These costs significantly influence resistance to new systems, as individuals often justify continued use of existing systems based on the time and effort already expended to master them (Tsai et al., 2019). In decision-making processes, sunk costs can distort cost–benefit evaluations, with individuals tending to favor choices consistent with their previous behaviors (Kulkov et al., 2023; Wang et al., 2018). Similarly, Tsai et al. (2019) found that sunk costs acted as a psychological barrier to the adoption of telehealth services. In the context of information systems, sunk costs may include cognitive, emotional, and procedural investments that are at risk of being forfeited when transitioning to a new platform (Petcharat et al., 2023).

In higher education, university students evaluating GAI-based learning systems may perceive sunk costs not only in financial terms but also in the time and cognitive effort previously invested in mastering existing learning strategies. These accumulated investments can strongly affect decision-making (Meireles et al., 2019; Santos et al., 2021). Emotional attachment to familiar systems (Dang et al., 2017; Liu et al., 2019), and a sense of competence and ownership, may further deter students from adopting new technologies, despite their potential benefits.

Drawing on Kashdan et al.'s (2004) finding that individuals high in curiosity and intrinsic motivation tend to invest more in learning, it is plausible that such individuals may also be more susceptible to the effects of sunk costs, although this relationship was not directly examined in their study. Deep engagement and extensive investment may lead to greater psychological commitment to existing systems (Pechenkina & Braniman, 2022). Although learners with high epistemic curiosity, particularly those driven by diverive or specific curiosity, tend to embrace novelty, they may still be affected by sunk costs. For instance, Kim and Lee (2017) found that both curiosity types were positively associated with perceived switching costs, including sunk costs, among mobile gamers. Their findings suggest that even highly curious individuals may hesitate to switch systems due to concerns about losing accumulated knowledge, social identity, or symbolic achievement. In such cases, curiosity may reinforce rather than reduce the impact of sunk costs, reflecting cognitive biases such as loss aversion and the endowment effect, which influence decision-making across educational contexts (Jordan & Duckett, 2018). In this study, sunk costs are conceptualized as learners' accumulated investments in an existing learning system or technology. Moreover, the abundance of alternatives may further heighten hesitation to adopt new platforms, particularly when learners perceive unclear advantages or anticipate a steep learning curve.

6 Privacy

Privacy is a critical concern in the adoption of new AI technologies (Zhang et al., 2025). It encompasses fears about the collection and storage of personal information, data usage without consent, misappropriation of sensitive details, and unauthorized sharing with third parties (Vimal Kumar et al., 2021). In the context of GAI, privacy concerns relate to the ethical handling and protection of users' data (Ashok et al., 2022; Kushwaha et al., 2022). This includes issues around how personal information is harvested, stored, and utilized by GAI systems. Privacy in GAI is defined by the protection against unauthorized data exploitation, ensuring informed consent, preventing the misuse of sensitive data, and avoiding unauthorized third-party access. Concerns also include the exposure of users' prior knowledge or input data during AI training, and the possibility of such data being used to fabricate content. The GAI system should provide safeguards against intrusions and unauthorized algorithmic access, particularly in the context of algorithm-driven decision-making (Acikgoz et al., 2023; Shin et al., 2022). In this study, privacy is conceptualized as the protection from intrusion or disruption of users' private information and knowledge while engaging with GAI systems.

7 Technology affordance

Affordance refers to what a thing is actually or anticipated to be used for, measuring the extent to which its attributes intuitively indicate usage (Fromm et al., 2020; Wu et al., 2022). Technology affordance specifically refers to the functions of technology that extend learning and perceptual capabilities, and these

functions may be economic, social, cognitive, or affective (Wu et al., 2022). Prior research has highlighted how the features of technology affordance influence the possibility, opportunity, and limitations for individuals choosing to engage with a system (Lawan et al., 2023). Technology affordances are a necessary condition for providing effective support in a constructivist online learning environment (Lee et al., 2023a, 2023b). GAI-based technology affordance refers to the use of GAI algorithm features to create conditions that support and enhance an individual's ability to perform tasks (Li & Wang, 2023; Nißen et al., 2022). In this study, technology affordance refers to GAI-based algorithm attributes such as explainability, accountability, and transparency (EAT), which are increasingly used to support students in gathering information and engaging in knowledge creation processes.

As the EAT of algorithms becomes increasingly important in current GIA systems, defining and operationalizing these concepts has become a critical task in both academia and industry (Balakrishnan et al., 2021; Chang et al., 2021; Haque et al., 2023; Shin et al., 2022). What once originated as a philosophical concern has evolved into a pressing issue in the humanities, particularly due to rapid changes in educational environments brought about by algorithmic advances (Shin & Park, 2019). The swift progression of GAI has opened new frontiers for learning and instruction while simultaneously raising concerns among educators, students, and policymakers (Celik, 2023). While educational institutions and stakeholders are actively engaged in addressing the social and ethical implications of these advancements, contentious debates remain, for example, how can we ensure the fairness of algorithms? What levels of transparency and accountability are acceptable? Universities must proactively address these concerns, as biased and non-transparent algorithms pose significant risks. It is imperative for higher education institutions to train students to differentiate between existing errors and potential risks in GAI systems.

8 Explainability

Concerns are growing regarding the explainability, accountability, and transparency of educational interventions supported by GAI systems (Memarian & Doleck, 2023; Steele, 2023; Yilmaz et al., 2023). One emerging approach to evaluating the value of GAI systems is to examine the transparency of their explanations and the rationale behind their decisions. Explainability refers to users' ability to understand the decision-making processes driven by GAI algorithms (Shin et al., 2022). User awareness and comprehension of how and why specific recommendations are made, and how user inputs influence those results, have been found to be critical (Shin et al., 2022). A reasonably explainable GAI system provides users with predictions that are easy to understand, thereby increasing users' acceptance of GAI systems.

Learning sources are often noisy, and reasoning through such data involves uncertainty at various levels (Kahneman et al., 2021). Prior studies have emphasized the importance of giving learners agency and responsibility for their own learning (Monzer et al., 2020; Pereira et al., 2021). Explainability supports learning by

revealing the rationale behind a GAI system's response, helping students refine their inquiries and improve their interactions with the system for better learning outcomes (Ma & Huo, 2023; Niu & Mvondo, 2024; Ratten & Jones, 2023). In the case of tools like ChatGPT, explainability refers to the extent to which the system can clearly communicate how it arrived at a specific response or decision. This is particularly relevant for students, who are often trained to understand the basis of the information provided to support learning, critical thinking, and research (Wu et al., 2022). Explainable GAI systems may offer insight into their reasoning processes, allowing students to assess the credibility and relevance of the information presented (Kaplan-Rakowski et al., 2023). When students understand how a GAI system generates responses, their trust in the system's reliability and utility tends to increase, which may enhance both acceptance and effective usage (Hyun & Kim, 2023; Koc et al., Ratten & Jones, 2023). However, students must also be aware of the potential biases and limitations of GAI systems to responsibly interpret and use the generated content (Bin-Nashwan et al., 2023; Dalalah & Dalalah, 2023; Ma & Huo, 2023).

9 Accountability

Accountability in algorithmic systems originates with the designers and developers responsible for their creation (Shin, 2020; Shin et al., 2022). Recognizing and preparing for potential unintended consequences is a fundamental aspect of ensuring algorithmic accountability. In the context of GAI, accountability refers to the responsibility to justify the actions, decisions, and outcomes produced by these systems (Kushwaha et al., 2022). It involves ensuring that these systems function fairly, ethically, and transparently, with mechanisms in place to address issues, risks, errors, or unintended consequences arising from their use (Memarian & Doleck, 2023; Shin et al., 2022). Accountability requires a clear delineation of responsibilities among all stakeholders involved in the development, deployment, and oversight of GAI systems, including designers, developers, users, and regulatory bodies. It is critical for fostering trust and ensuring the responsible integration of GAI across various domains (Kim et al., 2023; Omrani et al., 2022; Sun et al., 2023).

For learners, the presence of strong accountability measures can increase trust in GAI tools. This trust enhances their perception of usefulness, as they feel more confident in the reliability and integrity of the system's outputs. Accountable GAI systems are more likely to deliver reliable and relevant results for learners. When students know that a GAI system has been designed and is managed with accountability in mind, they are more likely to see it as a credible and effective tool for their educational tasks. Such awareness can lead to more targeted and productive use, thereby enriching the learning experience. GAI systems must also address complex ethical and privacy concerns, especially when handling personal data or intellectual property (Acikgoz et al., 2023; Kushwaha et al., 2022; Liu & Tao, 2022). Accountability ensures that GAI systems uphold ethical standards and protect user privacy (Priyambada et al., 2023; Sweeney, 2023; Vecchiarini & Somià, 2023). Learners' awareness of these safeguards may influence their perception of the system's trustworthiness and usefulness, particularly in contexts involving sensitive data.

10 Transparency

Transparency refers to the extent to which a GAI system appropriately discloses its influence on users' decision-making processes (Memarian & Doleck, 2023; Shin & Park, 2019). These systems should clearly indicate when their outputs significantly impact users and ensure responsible disclosure during user interaction (Memarian & Doleck, 2023; Shin, 2020). Transparency involves clarifying how such systems reason, manage data (particularly sensitive data), and what implicit knowledge they accumulate about users (Akter et al., 2021; Monzer et al., 2020). The “black box” nature of algorithms refers to the complexity and opacity of their inner workings, making it difficult for users to understand how decisions are made. Students using GAI systems to produce documents, complete assignments, or acquire new knowledge may unknowingly encounter or perpetuate societal biases embedded in the content generated by GAI algorithms (Akter et al., 2021). Therefore, GAI systems used in personalized learning must clearly communicate what data is collected, how that data informs content generation, and what safeguards are in place to ensure data privacy (Akter et al., 2021; Balakrishnan et al., 2021; Nißen et al., 2022).

Transparency is vital to building user trust (Gedrimiene et al., 2023). Bogina et al. (2021) emphasized that to foster an environment and ensure the fair and transparent processing of information, it is crucial for students and faculty who use GAI systems in learning, research, and administration to be fully informed about the sources of data. To promote transparent decision-making among learners, the functioning of GAI algorithms must be open to scrutiny. In practice, transparency involves providing systematic explanations that allow students to understand the data sources and reasoning behind GAI outputs. Such practices align with the pedagogical values of openness and integrity, and they empower students to critically and effectively engage with GAI technologies.

11 Perceived useful

Perceived usefulness refers to the degree to which an individual believes that utilizing specific information systems will enhance their performance (Lim et al., 2023). This definition indicates that perceived usefulness fundamentally relates to individual beliefs that influence decision-making. In the context of learners using GAI systems, perceived usefulness can be defined as the extent to which they believe that using a specific GAI system will enhance their learning performance and experience (Jain & Raghuram, 2024; Maheshwari, 2024). This belief extends beyond the system's functional capabilities and includes learners' evaluations of how the GAI system supports their educational activities (Vlachogianni & Tselios, 2022). If learners perceive that a GAI system provides accurate, relevant, and timely information or assistance, they are more likely to consider it useful for academic purposes. Currently, GAI systems offer extensive access to educational resources and information (Ratten & Jones, 2023; Vecchiarini & Somià, 2023). Learners may perceive these tools as useful if they deliver comprehensive and up-to-date knowledge that

enhances their academic pursuits (Kaplan- Rakowski et al., 2023). In essence, perceived usefulness among learners using GAI systems is shaped by subjective evaluations that positively influence their acceptance of GAI technologies (Zastudil et al., 2023).

12 Trust

Research by Slade et al. (2019) and Damgaard and Nielsen (2020) suggests a meaningful distinction between contextual factors (technological affordances) and dispositional antecedents (individual resistance factors) in technology-related decision-making. These studies indicate that trust primarily operates on perceptions of system-level characteristics rather than individual predispositions. Specifically, Damgaard and Nielsen (2020) found that phenomena such as sunk cost bias and SQB represent relatively stable cognitive heuristics that are less responsive to external modulating mechanisms such as trust. Similarly, Slade et al. (2019) observed that in educational technology contexts, trust is predominantly shaped by system attributes (such as transparency, consent procedures, and operational clarity) rather than by learners' pre-existing attitudes. Herdiani et al. (2024) further reinforce this perspective by highlighting trust as a critical mediating factor between system characteristics and user responses in educational environments involving sophisticated algorithmic systems. Their findings align with the present theoretical model, which positions trust as a moderating factor influencing how system features shape user awareness, rather than as a function of internal psychological constructs affecting cognitive evaluations.

In human–computer interaction, trust is a pivotal element (Choung et al., 2022). It involves understanding how users perceive, engage with, and depend on computer-based systems, directly impacting the effectiveness and acceptance of technological interfaces and solutions (Choung et al., 2022; John-Mathews, 2022; Omrani et al., 2022; Sun et al., 2023). Trust is typically defined as the confidence or belief in the reliability and credibility of GAI responses (Niu & Mvondo, 2024; Ratten & Jones, 2023). This includes users' assurance in the accuracy, dependability, and integrity of GAI outputs and interactions (Dalalah & Dalalah, 2023; Memarian & Doleck, 2023; Niu & Mvondo, 2024; Ratten & Jones, 2023). Previous studies have proposed that trust significantly influences the adoption of new technologies (Kim et al., 2023; Lee et al., 2023a, 2023b; Omrani et al., 2022). GAI represents an advanced tier of technology designed to emulate human intelligence and interact dynamically with educational environments (Ratten & Jones, 2023; Vecchiarini & Somià, 2023). The effectiveness of GAI integration in education depends largely on the trust learners place in these systems (Dalalah & Dalalah, 2023; Memarian & Doleck, 2023). Trust plays a crucial role in shaping the interface between students and GAI systems (Al Shamsi et al., 2022). Disparities in trust levels can lead to either underutilization or complete abandonment of these novel educational tools, underscoring the importance of fostering balanced trust that aligns with both the capabilities and limitations of GAI (Dalalah & Dalalah, 2023; Liu et al., 2023; Ratten & Jones, 2023).

13 Hypothesis development

13.1 Sunk cost and user awareness

In educational technology contexts, sunk cost traditionally signifies resistance to change due to prior investments (Arkes & Blumer, 1985). However, recent evidence suggests a more nuanced relationship. Damgaard and Nielsen (2020) found that users with significant investments often engage in more thorough information processing when evaluating alternatives, indicating heightened awareness rather than mere resistance. Similarly, Chen et al. (2024) observed that educators with substantial pedagogical commitments frequently demonstrate enhanced evaluations of new technologies. Cognitive engagement theory offers an explanatory framework for this phenomenon, wherein prior investment fosters sophisticated technological mental models that facilitate the evaluation of innovations (Santos et al., 2021). When encountering GAI systems, users with substantial investments may exhibit greater attentiveness to system features and applications due to their enhanced technological literacy and comparative evaluation skills (Balakrishnan et al., 2021). Accordingly, we propose:

H1: Sunk cost positively affects user awareness of adopting GAI systems.

14 Privacy and user awareness

Privacy concerns significantly impact AI technology adoption across various contexts. Choi (2021) identified widespread discomfort with AI-driven access to personal data, with users often unwilling to compromise privacy even for enhanced AI experiences. In higher education specifically, GAI tools such as ChatGPT have intensified concerns about managing sensitive student data (Moorhouse et al., 2023). Notably, privacy concerns may paradoxically enhance users' awareness of technology rather than merely inhibit adoption. When users recognize their data as valuable resources subject to extraction and monetization (Xin et al., 2022), this recognition can stimulate heightened attentiveness to system operations and data practices. Empirical evidence supports this dual relationship: while some studies report low concern when transparency is high (Vu et al., 2019), others document increased vigilance regarding data collection scope and limitations on user control (Liu & Khalil, 2023; Mutimukwe et al., 2022). This vigilance often manifests as increased information-seeking and critical evaluation of system features related to data handling. Students with elevated privacy concerns tend to more thoroughly examine GAI systems' data practices, thereby developing a more comprehensive awareness of system functionality and implications (Slade et al., 2019). This cognitive engagement extends beyond privacy features to broader system operations and capabilities. Accordingly, we propose the following hypothesis:

H2: Privacy concerns positively affect user awareness of adopting GAI systems.

15 Explainability and user awareness

Explainability refers to an AI system's capacity to articulate how and why specific outputs are generated in human-understandable terms (Khosravi et al., 2022). In educational contexts, this capability enhances interpretability, facilitates trust calibration, and supports learner autonomy (Kaplan-Rakowski et al., 2023; Rai, 2020). When GAI systems generate academic content without adequate explanatory mechanisms, users often struggle to evaluate output reliability or distinguish sound reasoning from algorithmic approximation (Khosravi et al., 2022). Conversely, when systems provide meta-information about their reasoning processes, users develop stronger cognitive frameworks for interpreting and evaluating outputs. This enhanced interpretability directly contributes to users' awareness of system capabilities, limitations, and appropriate applications. Explainable systems also enable metacognitive reflection, allowing learners to examine AI reasoning processes and enrich their understanding of both content and system operations. This cognitive engagement fundamentally expands users' awareness of when and how to effectively utilize GAI tools in educational contexts. Based on this evidence, we propose the following hypothesis:

H3: Explainability positively affects user awareness of adopting GAI systems.

16 Accountability and user awareness

In GAI contexts, accountability refers to systems operating in responsible, ethical, and justifiable ways regarding content generation and decision-making processes (Memarian & Doleck, 2023). In educational settings, this encompasses output accuracy, ethical compliance, fairness, and value alignment. GAI systems with robust accountability mechanisms including clear attribution, unbiased output generation, and traceable reasoning enhance user confidence and engagement. This engagement stimulates cognitive processing of system features and functions, thereby increasing user awareness. Research by Shin et al. (2022) and Bogina et al. (2021) demonstrates that accountable GAI tools foster deeper user understanding by necessitating critical evaluation of both informational quality and ethical dimensions. Furthermore, accountable systems support informed judgments by allowing users to trace output generation processes (Sahlgren, 2023; Yuan, 2023). This evaluative engagement activates metacognitive monitoring, expanding users' awareness of system capabilities and limitations. Ethical accountability features, such as educational value alignment and misuse prevention, have also been shown to encourage exploratory behavior and system monitoring (Celik, 2023; Niu & Mvondo, 2024), further enhancing user awareness. Based on this evidence, we propose the following:

H4: Accountability positively affects user awareness of adopting GAI systems.

17 Transparency and user awareness

Transparency in GAI systems refers to the disclosure of operational logic, data usage practices, and decision-making processes in an intelligible manner. In educational contexts, such transparency enables critical engagement with generated outputs and deeper understanding of system mechanisms (Gedrimiene et al., 2023; Memarian & Doleck, 2023). Research shows that transparency enhances user awareness through multiple cognitive pathways. When users gain insight into system operations, they develop more accurate mental models of AI functionality, which support more effective integration into learning strategies (Tayan et al., 2024). This understanding extends beyond basic functionality to encompass data handling practices, enhancing awareness of both technical processes and privacy implications (Dalalah & Dalalah, 2023; Ratten & Jones, 2023). Transparent systems also facilitate critical evaluation of content relevance, accuracy, and educational value, stimulating deeper cognitive processing and technological awareness (Memarian & Doleck, 2023). When data flows, model logic, and privacy safeguards are explicit, users show increased recognition of responsible AI-use principles (Bogina et al., 2021; Celik, 2023; Kong et al., 2023). This ethical awareness complements functional understanding, creating a comprehensive view of GAI capabilities and limitations. Based on this evidence, we propose the following:

H5: Transparency positively affects user awareness of adopting GAI systems.

18 User awareness and perceived usefulness

User awareness, defined as users' understanding, familiarity, and informed recognition of how GAI systems operate, plays a pivotal role in shaping perceptions of these technologies' educational value. In this context, perceived usefulness refers to the extent to which individuals believe the technology enhances teaching or learning performance. When users possess a comprehensive awareness of GAI functionalities, applications, and underlying mechanisms, they are more likely to appreciate its pedagogical benefits. For example, educators who understand GAI's capacity for personalized learning and content automation often view such tools as effective in improving instructional outcomes (Cheng et al., 2018; Klayklung et al., 2023). User awareness also extends to ethical and privacy-related dimensions. When users understand how GAI systems manage privacy risks and adhere to ethical standards, their confidence increases, enhancing perceived usefulness (Celik, 2023; Kong et al., 2023). Furthermore, awareness improves implementation quality: educators with deeper knowledge of GAI are better equipped to integrate these tools into instructional design, reinforcing their perceived value in educational practice (Liu et al., 2023). The literature suggests that user awareness contributes positively to evaluations of GAI usefulness, particularly by fostering informed adoption, trust, and pedagogical alignment. Therefore, we propose the following hypothesis:

H6: User awareness positively affects the perceived usefulness of GAI systems.

19 Trust as a moderating mechanism

In the context of GAI implementation, trust acts as a moderating mechanism between technological affordances and user awareness. The interaction becomes particularly salient in educational contexts, where students rely on GAI systems for learning and knowledge acquisition (Shahzad et al., 2024). The moderating role of trust is especially relevant in the relationship between explainability and user awareness (Al Shamsi et al., 2022). Kaplan-Rakowski et al. (2023) suggest that users with high trust in GAI systems are more likely to engage with system explanations and reasoning processes. This is supported by Hyun and Kim (2023), who found that trust enhances users' willingness to understand and apply system explanations. Similarly, Wu et al. (2022) demonstrate that trusted, explainable systems lead to greater comprehension and engagement.

H7a: Trust positively moderates the relationship between explainability and user awareness of GAI systems.

Trust also moderates the relationship between accountability and user awareness. Kushwaha et al. (2022) argue that trust in accountability measures leads users to critically engage with system features. Liu and Tao (2022) further found that trusted accountability mechanisms heighten users' attention to system capabilities, and Priyambada et al. (2023) similarly report improved awareness of system functionalities. These findings inform the following hypothesis:

H7b: Trust positively moderates the relationship between accountability and user awareness of GAI systems.

Transparency and trust likewise interact to influence user awareness. Bogina et al. (2021) highlight that transparent information processing fosters trust, shaping user perceptions of system features. Akter et al. (2021) found that trust enhances the effect of transparency on user understanding, while Nißen et al. (2022) confirm that trusted, transparent systems promote improved awareness and engagement. These understandings lead to the following hypothesis:

H7c: Trust positively moderates the relationship between transparency and user awareness of GAI systems.

Based on the preceding discussion, Fig. 1 below illustrates the hypothesized research model.

20 Research method

This study conducts a quantitative study based on survey data collected from target respondents with experience using GAI-based tools. The survey instrument comprised two distinct sections. The first section used nominal scales to capture demographic information and GAI usage patterns. The second section employed seven-point Likert scales (1 = strongly disagree to 7 = strongly agree) to measure theoretical constructs. All measurement items were adapted from validated scales in previous technology adoption studies, as detailed in Table 1 and Appendix A.

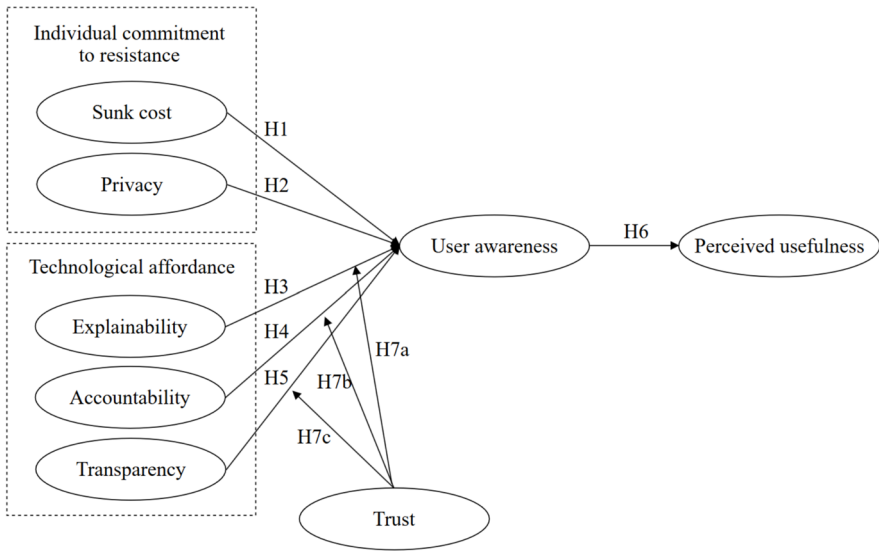


Fig. 1 Research model

21 Sample characteristics and data collection

A purposive sampling approach was adopted to examine user awareness of GAI in higher education, focusing on cognitive and technological affordance perspectives. To ensure accurate recall of interactions and perceived benefits, participation was limited to individuals who had used GAI tools within the past three months, following methodological guidelines established in prior technology adoption research (Wu et al., 2022).

Data was collected via a structured online survey administered over an eight-week period. To enhance representativeness and minimize selection bias, the survey was distributed across multiple digital platforms using a stratified random sampling approach across 24 daily time slots. A random reward system was implemented for 30 respondents to improve response rates while maintaining data quality.

The final sample comprised 572 undergraduate students from a comprehensive university in Taiwan. Gender distribution showed near parity, with female students representing 50.7% ($n = 290$) and male students 49.3% ($n = 282$). Age demographics indicated that 52.27% ($n = 299$) of participants were under 19 years old, 39.69% ($n = 227$) were between 19 and 22, and 8.04% ($n = 46$) were aged 20–29. This stratification aligns with typical undergraduate demographics in higher education, supporting the generalizability of the findings to similar academic contexts.

To ensure data quality and sample representativeness, several validation procedures were employed. Early and late respondents were compared using t-tests across key variables, revealing no significant differences ($p > 0.05$). Additionally, demographic characteristics were compared with university population statistics, confirming representative distribution across key indicators. These validation procedures

Table 1 Question items sources

Construct	Definitions and operational concepts	Items	Sources
Sunk cost	Occurs when people stick with endeavors in which they've already invested time, money, or other resources even when changing course would be the more logical choice	3	Popescu and Schut (2023)
Privacy	Protection intrusion and information access by algorithms in the context of algorithm driven decision making	4	Shin et al (2022)
Explainability	The availability of explanations about user decision making in relation to algorithms	4	Shin et al (2022)
Accountability	Principle about holding the designers and providers of algorithm-driven platforms liable for the consequences provided by their preset decision-making systems	4	Shin et al (2022)
Transparency	The algorithm-generated decisions made by AI should be open, verifiable, and/or visible to the users who use, adopt, and are affected by systems that employ such algorithms	4	Shin et al (2022)
Trust	Having confidence, faith, and/or hope in algorithms and algorithm-driven decision making	4	Fast and Jago (2020)
User awareness	The degree of understanding of GAI algorithms and how they could apply for learning purposes	4	Taylor and Brisini (2024)
Perceived usefulness	According to Davis (1989) refers to the degree to which an individual believes that using GAI services would improve the performance of his/her learning activities	4	Liu and Tao (2022)

enhance confidence in the sample's representativeness and the generalizability of the results. The methodological approach provides a robust foundation for investigating the relationships between individual commitment, technological affordance, and user awareness in the context of GAI adoption in higher education.

22 Measurement model assessment

The measurement instrument underwent rigorous validation through expert review involving three academic scholars who assessed the structure, content, and linguistic equivalence of the measures. Based on their feedback, refinements were made to enhance psychometric properties. Partial least squares structural equation modeling (PLS-SEM) was employed for data analysis due to its suitability for complex predictive models (Hair et al., 2019).

The *measurement model* was evaluated through reliability and validity assessments. Internal consistency was measured using Cronbach's alpha, while convergent and discriminant validity were assessed following the criteria established by Fornell and Larcker (1981). Item loadings ranged from 0.68 to 0.90, exceeding the recommended 0.60 threshold, indicating robust psychometric performance. Composite reliability values ranged from 0.86 to 0.93, surpassing the 0.70 benchmark, thus confirming strong internal consistency.

Average variance extracted (AVE) values ranged from 0.68 to 0.77, well above the 0.50 threshold, supporting convergent validity. Discriminant validity was confirmed using the Fornell-Larcker criterion: the square root of each construct's AVE exceeded its correlations with other constructs. Table 2 presents the psychometric properties of the measurement scales, and Table 3 shows discriminant validity through the correlation matrix and AVE square roots. Together, these findings confirm the strong psychometric robustness of the measurement model and provide a sound basis for structural model analysis. The established reliability and validity indicators suggest that the measures appropriately capture the intended constructs while maintaining distinctiveness among different theoretical dimensions.

Table 2 Descriptive statistics, reliability and convergent validity

Construct	Item means	Item loadings	AVE	CR	Cronbach's α
SC	4.73	0.68–0.90	0.68	0.86	0.76
PR	5.42	0.72–0.90	0.69	0.90	0.85
EX	5.00	0.85–0.89	0.76	0.93	0.90
AC	5.06	0.74–0.87	0.68	0.89	0.84
TR	5.12	0.80–0.87	0.69	0.90	0.85
UA	5.46	0.70–0.85	0.64	0.88	0.81
PU	5.56	0.83–0.90	0.76	0.93	0.89
TRU	4.86	0.87–0.89	0.77	0.93	0.90

Sunk cost (SC), Privacy (PR), Explainability (EX), Accountability (AC), Transparency (TR), User awareness(UA), Perceived usefulness (PU), Trust (TRU)

Table 3 Discriminant validity

	SC	PR	EX	AC	TR	UA	PU	TRU	Tru* TRA	Tru* AC	Tru* EX
SC	0.826										
PR	0.413	0.828									
EX	0.460	0.460	0.874								
AC	0.455	0.449	0.562	0.823							
TR	0.486	0.482	0.712	0.709	0.833						
UA	0.500	0.448	0.502	0.532	0.545	0.799					
PU	0.427	0.394	0.530	0.488	0.570	0.710	0.871				
TRU	0.446	0.268	0.511	0.541	0.665	0.447	0.479	0.879			
Tru*TRA	0.214	0.190	0.165	0.124	0.045	0.116	0.042	0.121	1.000		
Tru*AC	0.252	0.190	0.148	0.148	0.128	0.154	0.059	0.239	0.818	1.000	
Tru*EX	0.285	0.213	0.234	0.140	0.160	0.164	0.077	0.220	0.790	0.791	1.000

Diagonal value: Square root of AVE, Non-diagonal value: Correlation

PLS-SEM was used to test the *structural model* (Fig. 2). To assess its predictive capability, standardized path coefficients (β) were examined to evaluate hypothetical relationships, and statistical significance was assessed for each path. Coefficients of determination (R^2) were calculated for endogenous variables to determine the model's explanatory power.

23 Multi-group analysis (MGA) based on GAI tool usage experience

To address variations in GAI usage experiences among learners, a multi-group analysis (MGA) was conducted across three distinct GAI application contexts. Participants were classified based on their primary use of GAI tools: Group A used ChatGPT (for dialogue simulation, copywriting, logical reasoning, and language expression; $n = 256$); Group B used AI presentation tools (for structured slide design, automated formatting, and report generation; $n = 155$); and Group C employed text-to-image tools (for visual content creation, visual aids, and visualization tasks; $n = 161$). Before comparing path coefficients, measurement invariance was established using the MICOM procedure (Henseler et al., 2016) with 5,000 permutations and a significance level of $\alpha = 0.01$. This three-step process tested configural invariance (Step I), compositional invariance (Step II), and equality of means (Step IIIa) and variances (Step IIIb). Configural invariance (Step I) was established using identical indicators and data treatment across groups. Compositional invariance (Step II) was verified through permutation tests assessing construct correlations across groups. Steps IIIa and IIIb were then tested for equality of means and variances, respectively. Upon establishing measurement invariance, MGA was conducted using the Welch-Satterthwaite test to evaluate differences in structural path coefficients across the three GAI usage groups at the $\alpha = 0.01$ significance level.

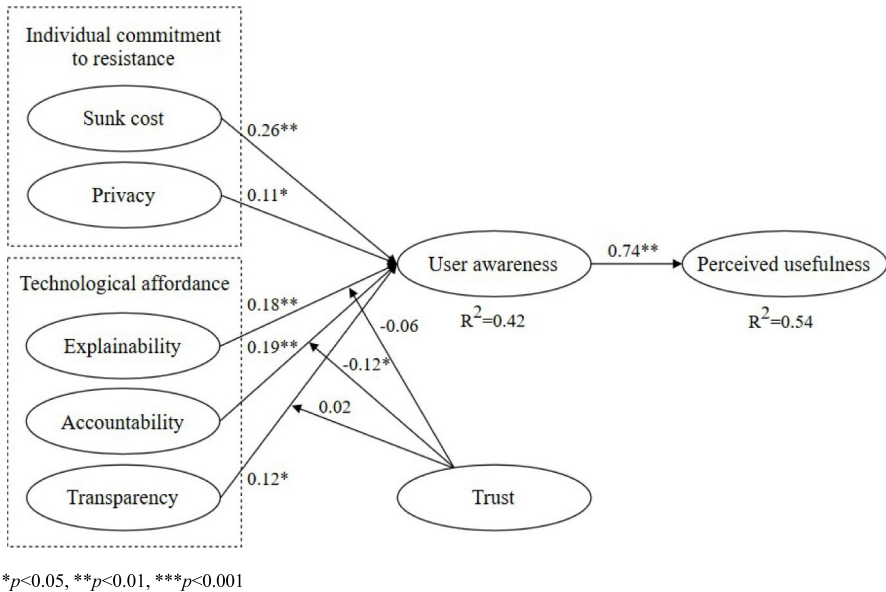


Fig. 2 Results of path analysis. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

24 Results

24.1 Descriptive analysis

Before examining the hypothesized relationships, descriptive statistics of the study constructs are presented. Table 2 shows that all constructs demonstrated mean values above the scale midpoint (4.0), indicating generally positive evaluations. Perceived usefulness (PU, Mean = 5.56) and user awareness (UA, Mean = 5.46) exhibited the highest means, suggesting that students have developed considerable awareness of GAI systems and perceive them as beneficial for their learning activities. Privacy concerns (PR, Mean = 5.42) also scored relatively high, reflecting students' growing sensitivity to data protection in algorithmic systems, consistent with Shin et al.'s (2022) findings on privacy concerns in algorithmic contexts. The technology affordance dimensions, transparency (TR, Mean = 5.12), accountability (AC, Mean = 5.06), and explainability (EX, Mean = 5.00), all demonstrated similar mean values, suggesting balanced perceptions of these system attributes. Trust (TRU, Mean = 4.86) and sunk cost (SC, Mean = 4.73) exhibited the lowest means, although still above the midpoint, indicating moderate levels of trust in GAI systems and acknowledgment of prior investments.

The Cronbach's alpha values ranged from 0.76 to 0.90, exceeding the recommended threshold of 0.70 (Hair et al., 2019), thus confirming internal consistency reliability. Convergent validity was established with AVE values ranging from 0.64 to 0.77, surpassing the 0.50 criterion (Fornell & Larcker, 1981). Discriminant validity was confirmed using the Fornell-Larcker criterion, as the square root of each construct's AVE exceeded its correlations with other constructs (Table 3).

25 Structural model results and hypothesis testing

Figure 2 presents the results of the structural model analysis using PLS-SEM. The model demonstrates strong explanatory power, accounting for 42% of the variance in user awareness (UA) and 54% in perceived usefulness (PU).

The results strongly support H1, revealing a significant positive relationship between sunk cost and user awareness ($\beta = 0.26$, $p < 0.01$). This indicates that students' prior investments in existing learning technologies enhance their awareness of GAI systems. H2 is also supported ($\beta = 0.11$, $p < 0.05$), confirming that privacy concerns positively influence students' awareness of GAI systems, although the effect is moderate compared to other predictors.

H3 is supported ($\beta = 0.18$, $p < 0.01$), demonstrating that explainability significantly enhances students' awareness of GAI systems. H4 is supported ($\beta = 0.19$, $p < 0.01$), confirming that accountability has a significant positive effect on user awareness, emerging as the strongest technological affordance factor influencing awareness. H5 is supported ($\beta = 0.12$, $p < 0.05$), indicating that transparency positively influences user awareness of GAI systems, although its effect is moderate compared to other technological affordance factors.

H6 is strongly supported ($\beta = 0.74$, $p < 0.01$), demonstrating a robust relationship between user awareness and perceived usefulness. This finding indicates that comprehensive awareness of GAI functionalities, applications, and underlying mechanisms significantly enhances students' appreciation of their pedagogical benefits.

The analysis also examined the moderating role of trust in the relationship between GAI system features and user awareness. H7a is not supported ($\beta = -0.06$, $p > 0.05$), indicating that trust does not significantly moderate the relationship between explainability and user awareness. This suggests that the effect of explainability on user awareness remains consistent regardless of trust levels. H7b is supported ($\beta = -0.12$, $p < 0.05$), confirming that trust significantly moderates the relationship between accountability and user awareness. However, the negative coefficient reveals that accountability's effect on user awareness is stronger when trust is lower, contrary to the initial prediction. H7c is not supported ($\beta = 0.02$, $p > 0.05$), indicating that trust does not significantly moderate the relationship between transparency and user awareness.

26 Multi-group analysis (MGA) results

The measurement invariance analysis using the MICOM procedure revealed varying levels of invariance across different group comparisons (Table 4). Full measurement invariance was confirmed between Groups A and B, with all constructs meeting the requirements for compositional invariance (Step II), equality of means (Step IIIa), and equality of variances (Step IIIb). For the A-C and B-C comparisons, compositional invariance (Step II) was established for all constructs, although full scalar invariance was not achieved, indicating partial

measurement invariance for these group pairs. As compositional invariance was established across all group comparisons, the minimum requirement for MGA was satisfied (Hair et al., 2019), allowing for meaningful comparison of path coefficients.

Following the establishment of measurement invariance, we conducted MGA using the Welch-Satterthwaite test to assess differences in structural path coefficients across the three GAI usage groups. Table 5 presents detailed MGA results. At the $\alpha = 0.01$ significance level, no statistically significant differences emerged between any pair of groups for any structural path.

The consistency of structural relationships across different GAI usage groups is particularly noteworthy given the distinct functional characteristics of these tools. Despite the varied nature of GAI applications, from text generation (ChatGPT) to presentation support to visual content creation, the core relationships among technological affordances, individual commitment to resistance, user awareness, and perceived usefulness remained remarkably stable. This finding suggests the robust generalizability of our theoretical framework across various GAI-enhanced learning environments, regardless of the specific tools employed. The MGA thus provides empirical support for the development of common strategic approaches to GAI implementation in educational settings, while acknowledging the functional diversity of these applications.

27 Discussion

27.1 Individual commitment factors and user awareness

The significant positive effect of sunk cost on user awareness (H1, $\beta = 0.26$, $p < 0.01$) aligns with Santos et al.'s (2021) cognitive engagement theory, which proposes that prior investments create sophisticated technological mental models that facilitate the evaluation of innovations. Students with substantial investments in current learning approaches demonstrate enhanced attentiveness to GAI features, consistent with Balakrishnan et al.'s (2021) observations regarding technology evaluation skills. This result extends Damgaard and Nielsen's (2020) finding that users with significant investments engage in more thorough information processing when evaluating alternatives. In our educational context, sunk cost emerged as the strongest predictor of user awareness, suggesting that students leverage their existing technological knowledge to develop a more comprehensive understanding of GAI systems.

The positive influence of privacy concerns on user awareness (H2, $\beta = 0.11$, $p < 0.05$) is consistent with Liu and Khalil's (2023) finding that privacy concerns in educational contexts stimulate heightened vigilance regarding technology features. Although the effect is moderate, this demonstrates that students with elevated privacy concerns engage in more thorough examinations of GAI systems, as documented by Slade et al. (2019). This finding also helps to explain the apparent paradox identified in the literature between privacy concerns and technology

Table 4 MICOM measurement

Construct	A and B Permutation (<i>p</i> -value)			A and C Permutation (<i>p</i> -value)			B and C Permutation (<i>p</i> -value)		
	Step II	Step IIIa		Step II	Step IIIa		Step II	Step IIIa	
		Step IIIb	Step IIIb		Step IIIb	Step IIIb		Step IIIb	Step IIIb
AC	0.100	0.637	0.629	0.550	0.002	0.223	0.349	0.003	0.509
EX	0.579	0.451	0.291	0.432	0.004	0.030	0.340	0.010	0.137
PR	0.927	0.026	0.601	0.804	0.008	0.321	0.442	0.564	0.168
PU	0.026	0.936	0.622	0.444	0.446	0.726	0.397	0.360	0.403
SC	0.032	0.432	0.619	0.169	0.152	0.825	0.208	0.019	0.322
TRA	0.029	0.143	0.678	0.507	0.003	0.049	0.286	0.095	0.001
Tru	0.456	0.024	0.094	0.473	0.000	0.001	0.883	0.007	0.042
UA	0.162	0.189	0.336	0.076	0.515	0.587	0.828	0.032	0.986
Conclusion	Total invariance			Partial invariance			Partial invariance		

Note:

- Significance level $\alpha = 0.01$
- Step II tests Compositional Invariance; Step IIIa tests Equality of Means; Step IIIb tests Equality of Variances

Table 5 Multi-group analysis (MGA) results

Hypotheses	Original path coefficient			Welch-Satterthwaite test		
	β (A)	β (B)	β (C)	<i>p</i> -value (A vs B)	<i>p</i> -value (A vs C)	<i>p</i> -value (B vs C)
AC—> UA	0.182	0.203	0.055	0.890	0.326	0.381
EX—> UA	−0.008	0.322	0.101	0.032	0.434	0.129
PR—> UA	0.327	0.110	0.167	0.108	0.214	0.706
SC—> UA	0.081	0.205	0.397	0.270	0.018	0.207
TRA—> UA	0.101	−0.115	0.219	0.318	0.375	0.118
Tru—> UA	0.215	0.188	−0.020	0.840	0.033	0.140
UA—> PU	0.780	0.673	0.718	0.087	0.383	0.600
Tru x TRA—> UA	−0.047	0.171	−0.016	0.193	0.817	0.268
Tru x EX—> UA	−0.192	0.006	0.031	0.256	0.105	0.890
Tru x AC—> UA	0.161	−0.104	−0.036	0.141	0.134	0.699

Note:

- Significance level $\alpha=0.01$
- β =Standardized path coefficient for each group

engagement. Rather than simply inhibiting adoption, as suggested by traditional models, our results show that privacy concerns can function as cognitive triggers that enhance overall system awareness, providing empirical support for Xin et al.'s (2022) proposition that recognizing data as valuable resources can stimulate attentiveness to system operations.

28 Technological affordance factors and user awareness

The significant effect of explainability on user awareness (H3, $\beta=0.18$, $p<0.01$) aligns with Khosravi et al.'s (2022) assertion that explanatory mechanisms improve system interpretability in educational settings. When GAI systems provide meta-information about their reasoning processes, students develop stronger cognitive frameworks for interpreting and evaluating outputs. This results in enhanced user awareness, supporting Kaplan-Rakowski et al.'s (2023) finding that explainable AI facilitates trust calibration and learner autonomy. In the university context, explainability appears to be particularly important for developing a comprehensive awareness of system capabilities and limitations, as students value understanding the underlying logic of AI-generated content for their academic tasks.

The finding that accountability has a significant positive effect on user awareness (H4, $\beta=0.19$, $p<0.01$) aligns with Memarian and Doleck's (2023) emphasis on the importance of responsible and ethical operation of GAI in educational contexts. This supports research by Shin et al. (2022) and Bogina et al. (2021), which shows that accountable GAI tools foster deeper user understanding through critical evaluation of both informational quality and ethical dimensions. The strong effect of accountability suggests that university students particularly value systems that demonstrate

clear responsibility for the content they generate, consistent with Yuan's (2023) and Sahlgren's (2023) findings that accountable systems enable users to make informed judgments about reliability and appropriateness.

The positive influence of transparency on user awareness ($H5$, $\beta = 0.12$, $p < 0.05$) aligns with theoretical propositions by Gedrimiene et al. (2023) and Memarian and Doleck (2023) that transparency enables critical engagement with GAI systems. Although the effect is moderate compared to other technological affordance factors, it supports Tayan et al.'s (2024) assertion that transparent systems help users develop more accurate mental models of AI functionality. In the university context, transparency appears to facilitate understanding of both technical processes and privacy implications, consistent with Dalalah and Dalalah's (2023) research on the role of transparency in educational AI settings.

29 User awareness and perceived usefulness

The robust relationship between user awareness and perceived usefulness ($H6$, $\beta = 0.74$, $p < 0.01$) provides substantial empirical validation for Klayklung et al.'s (2023) and Cheng et al.'s (2018) theoretical propositions regarding the importance of awareness in shaping perceptions of educational technology. The strength of this relationship suggests that comprehensive awareness of GAI functionalities, applications, and underlying mechanisms significantly enhances students' appreciation of its pedagogical benefits. This alignment supports Liu et al.'s (2023) finding that awareness contributes to perceived usefulness through informed adoption and trust. The strong effect size indicates that user awareness serves as a critical determinant of GAI acceptance in educational settings, extending Celik's (2023) and Kong et al.'s (2023) research on awareness of ethical and privacy-related dimensions in educational AI contexts.

30 Moderating effects of trust

The unexpected non-significant moderating effect of trust on the relationship between explainability and user awareness ($H7a$, $\beta = -0.06$, $p > 0.05$) can be attributed to several theoretical and contextual factors. In the higher education context, students may approach GAI systems with an inherently critical and analytical mindset, focusing primarily on understanding system explanations regardless of their trust levels. As Kaplan-Rakowski et al. (2023) suggest, students in academic settings tend to prioritize comprehension of system functionality over trust-based interactions. Additionally, the complexity of GAI explanations may create a direct cognitive processing pathway that operates independently of trust considerations. Wu et al. (2022) note that when students engage with complex technological explanations, their cognitive resources may be predominantly focused on understanding the content rather than being influenced by trust perceptions. This cognitive load theory perspective suggests that the mental effort required to process GAI explanations may overshadow the potential moderating effect of trust.

The significant negative moderating effect of trust on the relationship between accountability and user awareness (H7b, $\beta = -0.12$, $p < 0.05$) suggests a compensatory mechanism in which users with lower trust levels scrutinize accountability features more carefully, potentially to validate system reliability. While this contradicts aspects of Kushwaha et al.'s (2022) framework, it aligns with emerging perspectives in technology adoption literature suggesting that skeptical users may engage more critically with accountability mechanisms (Liu & Tao, 2022). This finding reveals an important dynamic in educational technology adoption: when students distrust a system, they become more attentive to its accountability features, suggesting that accountability may function as a trust-building mechanism rather than being enhanced by pre-existing trust.

The non-significant moderating effect of trust on the relationship between transparency and user awareness (H7c, $\beta = 0.02$, $p > 0.05$) might be explained by transparency functioning as a fundamental system requirement rather than a trust-building mechanism in higher education settings. As Bogina et al. (2021) argue, when information transparency is viewed as an inherent system characteristic, its relationship with user awareness may operate independently of trust levels. Additionally, the technical complexity of GAI system transparency may influence students' cognitive processing patterns. When system transparency involves intricate technical details, students may focus primarily on direct information comprehension rather than engaging in trust-mediated processing (Nißen et al., 2022).

31 Generalizability across GAI tool types

Beyond analyzing specific hypothesized relationships, we examined the consistency of our model across diverse GAI usage contexts through MGA. Participants were classified into three distinct groups based on their primary GAI tool usage. Our analysis revealed remarkable consistency in structural relationships across these functionally distinct GAI applications. Despite the varied nature of these tools, the relationships between technological affordances, individual commitment to resistance, user awareness, and perceived usefulness retained their significance and relative strength across all groups. This structural invariance is noteworthy given that these GAI tools serve fundamentally different purposes and engage users through distinct interaction modes.

This finding aligns with Wu et al.'s (2022) research on technology affordance, which demonstrated that technological features could effectively support learning processes and facilitate learning outcomes. Our study extends this theoretical understanding by providing evidence that relationship patterns between technological characteristics and user perceptions remain stable even across functionally diverse GAI applications. This suggests that the fundamental attributes of GAI technologies (such as explainability, transparency, and accountability) may exert greater influence on user perceptions and usage behavior than application-specific functionalities. Such consistency in structural relationships underscores the value of focusing on these core GAI attributes when designing educational implementations, as they appear to function similarly across various learning contexts and user populations.

32 Theoretical contributions

Our study makes several substantial theoretical contributions to the literature on technology adoption in educational contexts. First, it advances understanding of GAI adoption by revealing the nuanced mechanisms between individual commitment factors and technological affordances. The findings demonstrate that sunk cost ($\beta = 0.26$, $p < 0.01$) significantly influences user awareness, extending previous technology acceptance models by highlighting the crucial role of psychological investment in evaluating new technologies. This relationship challenges traditional views of sunk cost as merely a resistance factor, suggesting instead that prior technological investments foster sophisticated mental models that enhance recognition and evaluation of new system features. This contribution refines our understanding of how individual commitment factors shape technology awareness and acceptance in educational contexts.

Second, this study enhances understanding of the hierarchical importance of technological affordance features in educational settings. Empirical evidence reveals that accountability ($\beta = 0.19$, $p < 0.01$) and explainability ($\beta = 0.18$, $p < 0.01$) exert stronger influences than transparency ($\beta = 0.12$, $p < 0.05$) on user awareness. These findings challenge the conventional assumption that all technological affordances hold equal weight in adoption decisions, suggesting a more nuanced understanding of how specific system features contribute to user awareness. This pattern provides theoretical insights for feature prioritization in educational GAI development and extends affordance theory by demonstrating how different affordance dimensions function in complex technological environments. This contribution is particularly relevant given the increasing emphasis on responsible AI in education, where understanding the relative importance of system features is crucial for theory development.

Third, this study significantly contributes to trust theory in technology adoption by revealing the complex and differential moderating role of trust in GAI adoption. The unexpected negative moderating effect between trust and accountability on user awareness ($\beta = -0.12$, $p < 0.05$) suggests a compensatory mechanism that contradicts traditional assumptions about trust's universally positive role. Notably, trust significantly moderates the relationship with accountability, but not explainability or transparency. This finding proposes a more nuanced theoretical framework in which trust dynamics depend on specific system features and functional contexts. It extends trust theory by suggesting that moderate skepticism may promote more thorough engagement with features related to accountability and responsibility, adding depth to existing models of trust in educational technology.

Fourth, the MGA demonstrates remarkable consistency in structural relationships across functionally diverse GAI applications, offering a significant contribution to understanding technological adoption patterns. This cross-context consistency suggests that psychological mechanisms driving GAI adoption operate at a level that transcends specific tool functionalities, supporting the development of generalizable frameworks. The structural invariance observed across tools that serve fundamentally different purposes and interaction modalities provides empirical support for the existence of shared adoption mechanisms. This contributes to more comprehensive theoretical models that can accommodate the diversity of emerging GAI

applications and extends Wu et al.'s (2022) work by demonstrating the stability of core relationship patterns in the evolving educational technology landscape.

Fifth, our mediation analysis identifies user awareness as a critical intermediate mechanism, contributing to adoption theory by clarifying how individual commitment and technological affordance factors influence students' perceptions of system usefulness. This mediating role explains how initial attitudes and understanding of system features translate into evaluations of GAI's utility in academic contexts, building on Vlachogianni and Tselios's (2022) findings. The strong mediation effect ($\beta = 0.74$, $p < 0.01$) underscores the importance of awareness as a cognitive precursor to perceived usefulness, suggesting a sequential rather than parallel adoption process. This insight provides a more precise theoretical framework for explaining how individual and technological factors interact to shape GAI acceptance and suggests new avenues for future research on awareness-building in technology adoption.

33 Practical implications

The findings offer valuable practical implications for stakeholders in educational technology. First, the strong relationship between user awareness and perceived usefulness ($\beta = 0.74$, $p < 0.01$) indicates that institutions should implement training programs that enhance students' understanding of GAI capabilities and limitations. These programs should emphasize the link between system features and learning outcomes to support informed adoption. Institutions are advised to prioritize initiatives that build understanding of both functional and ethical aspects of GAI systems through workshops, demonstrations, and clear documentation highlighting explainability and accountability to promote engagement and acceptance.

Second, the differential impacts of technological affordances provide actionable guidance for developers and vendors. The hierarchical effects of accountability ($\beta = 0.19$, $p < 0.01$), explainability ($\beta = 0.18$, $p < 0.01$), and transparency ($\beta = 0.12$, $p < 0.05$) suggest that system design should prioritize features that enhance accountability and explainability. Developers should implement mechanisms that clearly communicate content responsibility and system logic, with these priorities reflected in product roadmaps and marketing to educational clients.

Third, findings related to privacy concerns ($\beta = 0.11$, $p < 0.05$) and trust dynamics have implications for institutional and regulatory policy. Institutions should establish data protection policies and transparent usage guidelines that address privacy while fostering trust. Recognizing that privacy concerns may enhance rather than inhibit awareness, institutions should also encourage critical engagement with GAI outputs, particularly in academic contexts where critical evaluation is valued. This can be supported through curriculum integration that emphasizes critical assessment alongside productive system use.

Fourth, the cross-application consistency found in our MGA suggests that institutions can develop broadly applicable GAI implementation strategies. Rather than customizing approaches for each tool, common frameworks for enhancing awareness and perceived usefulness may be adopted across tools. This enables efficient resource use while still encouraging instructional designers to account for the unique affordances of specific applications within learning activities. This balanced

approach respects the functional diversity of GAI tools while maximizing administrative efficiency in technology implementation.

Fifth, our findings regarding the differential moderating effects of trust have important implications for user interface design and communication strategies. The negative moderation between trust and accountability ($\beta = -0.12$, $p < 0.05$) suggests that system designers should implement different information presentation strategies based on user trust levels. Low-trust users may benefit from prominent accountability information, while high-trust users might respond better to other features. Adaptive interfaces responsive to user trust signals could enhance GAI effectiveness. Moreover, since trust does not moderate explainability and transparency, these features should be consistently presented, as their influence appears to operate through direct cognitive pathways.

Sixth, based on our mediation analysis, we recommend staged implementation strategies that recognize user awareness as a critical intermediate step. Institutions should begin with initiatives that build understanding of system capabilities and limitations before promoting adoption benefits. Progressive training modules, starting with foundational knowledge and advancing to more complex applications, align with our finding that awareness mediates the effects of commitment and affordance on perceived usefulness.

34 Limitations and future research

This study opens several avenues for future research. The cross-sectional nature of our data limits causal inference, suggesting a need for longitudinal studies to examine how the identified relationships evolve as students gain greater experience with GAI tools. While this study focused on three key affordances, emerging systems may introduce additional features worth exploring. Future research could also examine potential three-way interactions involving system features, user experience, or technological self-efficacy to deepen understanding of the conditions under which trust effectively moderates adoption in educational contexts.

Appendix

Demographic Information

Age, gender, academic level, field of study, primary GAI tool usage, primary usage purpose

Survey Items

All items were measured using a 7-point Likert scale ranging from 1 = Strongly disagree to

7 = Strongly agree.

Sunk Cost (SC)

SC1. When learning to use generative AI tools, I had already spent a lot of time learning how to use existing technologies and their functions.

SC2. When learning to use generative AI tools, I had already spent a lot of time becoming proficient in using existing technologies and their functions.

SC3. Even if generative AI tools may be more effective for learning, I still tend to use the tools and methods I am familiar with.

Privacy (PR)

PR1. When using generative AI tools, I feel highly uncertain about providing personal information to the platform.

PR2. When using generative AI tools, disclosing personal information to the platform may involve high risks and potential losses.

PR3. When using generative AI tools, I worry that my personal information may be used without permission.

PR4. When using generative AI tools, I handle personal information with caution to protect my privacy.

Explainability (EX)

EX1. The generative AI tools I frequently use allow me to understand their decision-making and reasoning processes.

EX2. The generative AI tools I frequently use help me understand the rationale behind their suggestions.

EX3. The generative AI tools I frequently use allow me to understand how they generate content.

EX4. The computational processes of the generative AI tools I frequently use are transparent and explainable to me.

Accountability (AC)

AC1. The generative AI tools I frequently use require developers to be held accountable for harmful or inappropriate content.

AC2. The generative AI tools I frequently use should be governed by regulatory mechanisms to ensure responsible use.

AC3. The generative AI tools I frequently use allow access to usage policies and guidelines.

AC4. The generative AI tools I frequently use permit third-party auditing of their algorithms.

Transparency (TR)

TR1. The generative AI tools I frequently use provide easy access to relevant information and documentation.

TR2. The generative AI tools I frequently use help me understand the reasons behind the generated outputs.

TR3. The decision criteria and evaluation mechanisms of the generative AI tools I frequently use are open and transparent to users.

TR4. Even for non-technical users, the internal workings of the generative AI tools I frequently use are generally understandable.

Trust (TRU)

TRU1. I trust the information provided by generative AI platforms.

TRU2. The results generated by generative AI are trustworthy.

TRU3. Even if I do not fully understand how they work, I trust generative AI tools to produce accurate outputs.

TRU4. Overall, I have confidence in the results produced by generative AI tools.

User Awareness (UA)

UA1. When using generative AI tools, I am aware of their functions and potential benefits for learning.

UA2. When using generative AI tools, I explore their features and adapt my usage to enhance learning outcomes.

UA3. When using generative AI tools, I am aware of the potential risks and challenges involved.

UA4. When using generative AI tools, I remain attentive and understand the outputs generated.

Perceived Usefulness (PU)

PU1. Using generative AI tools improves my learning performance.

PU2. Using Generative AI tools increases my productivity for my learning.

PU3. Generative AI tools provide accurate and timely information or assistance that is useful for my learning.

PU4. Generative AI tools offer extensive educational resources and knowledge that enhance my learning experience.

Acknowledgements This research was financially supported by the National Science and Technology Council, Taiwan, under Grant No. 113-2410-H-328-004.

Data availability Data collected from this study is available from the corresponding author on reasonable request.

Declaration

Competing interests The author declares that she has no conflict of interest that is directly or indirectly related to the work submitted for publication.

References

- Acikgoz, F., Perez-Vega, R., Okumus, F., & Stylos, N. (2023). Consumer engagement with AI-powered voice assistants: A behavioral reasoning perspective. *Psychology & Marketing*, 40(11), 2226–2243. <https://doi.org/10.1002/mar.21873>
- Ajzen, I. (2002). Perceived behavioral control, self-efficacy, locus of control, and the theory of planned behavior 1. *Journal of Applied Social Psychology*, 32(4), 665–683. <https://doi.org/10.1111/j.1559-1816.2002.tb00236.x>
- Akter, S., McCarthy, G., Sajib, S., Michael, K., Dwivedi, Y. K., D'Ambra, J., & Shen, K. N. (2021). Algorithmic bias in data-driven innovation in the age of AI. *International Journal of Information Management*, 60. <https://doi.org/10.1016/j.ijinfomgt.2021.102387>
- Al Shamsi, J. H., Al-Emran, M., & Shaalan, K. (2022). Understanding key drivers affecting students' use of artificial intelligence-based voice assistants. *Education and Information Technologies*, 27(6), 8071–8091. <https://doi.org/10.1007/s10639-022-10947-3>
- Arkes, H. R., & Blumer, C. (1985). The psychology of sunk cost. *Organizational Behavior and Human Decision Processes*, 35(1), 124–140. [https://doi.org/10.1016/0749-5978\(85\)90049-4](https://doi.org/10.1016/0749-5978(85)90049-4)
- Ashok, M., Madan, R., Joha, A., & Sivarajah, U. (2022). Ethical framework for Artificial Intelligence and Digital technologies. *International Journal of Information Management*, 62. <https://doi.org/10.1016/j.ijinfomgt.2021.102433>
- Balakrishnan, J., Dwivedi, Y. K., Hughes, L., & Boy, F. (2021). Enablers and Inhibitors of AI-Powered Voice Assistants: A Dual-Factor Approach by Integrating the Status Quo Bias and Technology Acceptance Model. *Information Systems Frontiers*, 26, 921–942. <https://doi.org/10.1007/s10796-021-10203-y>

- Bilquise, G., Ibrahim, S., & Salhie, S. E. M. (2024). Investigating student acceptance of an academic advising chatbot in higher education institutions. *Education and Information Technologies*, 29(5), 6357–6382. <https://doi.org/10.1007/s10639-023-12076-x>
- Bin-Nashwan, S. A., Sadallah, M., & Boutera, M. (2023). Use of ChatGPT in academia: Academic integrity hangs in the balance. *Technology in Society*, 75. <https://doi.org/10.1016/j.techsoc.2023.102370>
- Bogina, V., Hartman, A., Kuflik, T., & Shulner Tal, A. (2021). Educating software and AI stakeholders about algorithmic fairness, accountability, transparency and ethics. *International Journal of Artificial Intelligence in Education*, 32, 808–833. <https://doi.org/10.1007/s40593-021-00248-0>
- Celik, I. (2023). Towards Intelligent-TPACK: An empirical study on teachers' professional knowledge to ethically integrate artificial intelligence (AI)-based tools into education. *Computers in Human Behavior*, 138. <https://doi.org/10.1016/j.chb.2022.107468>
- Chang, Y., Lee, S., Wong, S. F., & Jeong, S. P. (2021). AI-powered learning application use and gratification: An integrative model. *Information Technology & People*, 35(7), 2115–2139. <https://doi.org/10.1108/itp-09-2020-0632>
- Chen, C. Q., Wang, C.-Y., Shan, X. F., Zhan, L., & Chen, S. J. (2024). An empirical investigation of reasons influencing pre-service teachers acceptance and rejection of immersive virtual reality usage. *Teaching and Teacher Education*, 137, 104391. <https://doi.org/10.1016/j.tate.2023.104391>
- Cheng, Y. W., Sun, P. C., & Chen, N. S. (2018). The essential applications of educational robot: Requirement analysis from the perspectives of experts, researchers and instructors. *Computers & Education*, 126, 399–416. <https://doi.org/10.1016/j.compedu.2018.07.020>
- Choi, Y. (2021). A study of employee acceptance of artificial intelligence technology. *European Journal of Management and Business Economics*, 30(3), 318–330. <https://doi.org/10.1108/ejmb-06-2020-0158>
- Choung, H., David, P., & Ross, A. (2022). Trust in AI and Its Role in the Acceptance of AI Technologies. *International Journal of Human-Computer Interaction*, 39(9), 1727–1739. <https://doi.org/10.1080/10447318.2022.2050543>
- Dalalah, D., & Dalalah, O. M. (2023). The false positives and false negatives of generative AI detection tools in education and academic research: The case of ChatGPT. *The International Journal of Management Education*, 21(2). <https://doi.org/10.1016/j.ijme.2023.100822>
- Damgaard, M. T., & Nielsen, H. S. (2020). Behavioral economics and nudging in education: Evidence from the field. In *The Economics of Education* (pp. 21–35). Academic Press. <https://doi.org/10.1016/B978-0-12-815391-8.00002-1>
- Dang, Y. M., Zhang, Y. G., & Morgan, J. N. (2017). Integrating switching costs to information systems adoption: An empirical study on learning management systems. *Information Systems Frontiers*, 19, 625–644. <https://doi.org/10.1007/S10796-015-9618-6>
- Davis, F. D., Bagozzi, R. P., & Warshaw, P. R. (1989). User Acceptance of Computer Technology: A Comparison of Two Theoretical Models. *Management Science*, 35(8), 982–1003. <https://doi.org/10.1287/mnsc.35.8.982>
- Duong, C. D., Vu, T. N., & Ngo, T. V. (2023). Applying a modified technology acceptance model to explain higher education students' usage of ChatGPT: A serial multiple mediation model with knowledge sharing as a moderator. *The International Journal of Management Education*, 21(3). <https://doi.org/10.1016/j.ijme.2023.100883>
- Fast, N. J., & Jago, A. S. (2020). Privacy matters... or does It? Algorithms, rationalization, and the erosion of concern for privacy. *Current Opinion in Psychology*, 31, 44–48. <https://doi.org/10.1016/j.copsyc.2019.07.011>
- Fornell, C., & Larcker, D. F. (1981). Evaluating structural equation models with unobservable variables and measurement error. *Journal of Marketing Research*, 18(1), 39–50. <https://doi.org/10.1177/002224378101800104>
- Fromm, J., Mirbabaie, M., & Stieglitz, S. (2020). A Systematic Review of Empirical Affordance studies: Recommendations for Affordance Research in Information Systems. *Twenty-Eighth European Conference on Information Systems (ECIS2020)*, Marrakesh, Morocco.
- Gedrimiene, E., Celik, I., Mäkitalo, K., & Muukkonen, H. (2023). Transparency and trustworthiness in user intentions to follow career recommendations from a learning analytics tool. *Journal of Learning Analytics*, 10(1), 54–70. <https://doi.org/10.18608/jla.2021.7791>
- Hair, J. F., Black, W. C., Babin, B. J., & Anderson, R. E. (2019). Multivariate data analysis. Cengage Learning EMEA.

- Haque, A. K., Islam, A. K., & Mikalef, P. (2023). Explainable Artificial Intelligence (XAI) from a user perspective: A synthesis of prior literature and problematizing avenues for future research. *Technological Forecasting and Social Change*, 186. <https://doi.org/10.1016/j.techfore.2022.122120>
- Henseler, J., Hubona, G., & Ray, P. A. (2016). Using PLS path modeling in new technology research: Updated guidelines. *Industrial Management & Data Systems*, 116(1), 2–20. <https://doi.org/10.1108/IMDS-09-2015-0382>
- Herdiani, A., Mahayana, D., & Rosmansyah, Y. (2024). Building Trust in an Artificial Intelligence-Based Educational Support System: A Narrative Review. *Jurnal Sosioteknologi*, 23(1), 101–119. <https://doi.org/10.5614/sostek.itbj.2024.23.1.6>
- Hung, C. M. (2023). Semiconductor Chip Design in a Legoland. 2023 *IEEE Asian Solid-State Circuits Conference (A-SSCC)*. 1–4, Haikou, China. <https://doi.org/10.1109/A-SSCC58667.2023.10347923>
- Hyun Baek, T., & Kim, M. (2023). Is ChatGPT scary good? How user motivations affect creepiness and trust in generative artificial intelligence. *Telematics and Informatics*, 83, 102030. <https://doi.org/10.1016/j.tele.2023.102030>
- Jain, K. K., & Raghuram, J. N. V. (2024). Gen-AI integration in higher education: Predicting intentions using SEM-ANN approach. *Education and Information Technologies*, 29(13), 1–41. <https://doi.org/10.1007/s10639-024-12506-4>
- John-Mathews, J. M. (2022). Some critical and ethical perspectives on the empirical turn of AI interpretability. *Technological Forecasting and Social Change*, 174. <https://doi.org/10.1016/j.techfore.2021.121209>
- Jordan, M. M., & Duckett, N. D. (2018). Universities confront ‘tech disruption’: Perceptions of student engagement online using two learning management systems. *The Journal of Public and Professional Sociology*, 10(1), 4. <https://doi.org/10.62915/2154-8935.1135>
- Kahneman, D., Sibony, O., & Sunstein, C. R. (2021). *Noise: a flaw in human judgment*: Hachette UK.
- Kaplan-Rakowski, R., Grotewold, K., Hartwick, P., & Papin, K. (2023). Generative AI and teachers’ perspectives on its implementation in education. *Journal of Interactive Learning Research*, 34(2), 313–338.
- Kashdan, T. B., Rose, P., & Fincham, F. D. (2004). Curiosity and exploration: Facilitating positive subjective experiences and personal growth opportunities. *Journal of Personality Assessment*, 82(3), 291–305. https://doi.org/10.1207/s15327752jpa8203_05
- Khosravi, H., Shum, S. B., Chen, G., Conati, C., Tsai, Y.S., Kay, J., . . . Gašević, D. (2022). Explainable Artificial Intelligence in education. *Computers and Education: Artificial Intelligence*, 3. <https://doi.org/10.1016/j.caeai.2022.100074>
- Kim, Y. B., & Lee, S. H. (2017). Mobile gamer’s epistemic curiosity affecting continuous play intention: Focused on players’ switching costs and epistemic curiosity. *Computers in Human Behavior*, 77, 32–46. <https://doi.org/10.1016/j.chb.2017.08.023>
- Kim, Y., Seok, J., & Roh, T. (2023). The linkage between quality of information systems and the impact of trust-based privacy on behavioral outcomes in unmanned convenience store: Moderating effect of gender and experience. *Technological Forecasting and Social Change*, 196. <https://doi.org/10.1016/j.techfore.2023.122852>
- Klayklung, P., Chocksathaporn, P., Limna, P., Kraiwanit, T., & Jangjarat, K. (2023). Revolutionizing Education with ChatGPT: Enhancing Learning Through Conversational AI. *Universal Journal of Educational Research*, 2(3), 217–225.
- Koc, E., Hatipoglu, S., Kivrak, O., Celik, C., & Koc, K. (2023). Houston, we have a problem!: The use of ChatGPT in responding to customer complaints. *Technology in Society*, 74, 102333. <https://doi.org/10.1016/j.techsoc.2023.102333>
- Kong, S. C., Cheung, W. M. Y., & Zhang, G. (2023). Evaluating an artificial intelligence literacy programme for developing university students’ conceptual understanding, literacy, empowerment and ethical awareness. *Educational Technology & Society*, 26(1), 16–30. <https://doi.org/10.1016/j.chbr.2022.100223>
- Kulkov, I., Kulkova, J., Rohrbeck, R., Menvielle, L., Kaartemo, V., & Makkonen, H. (2023). Artificial intelligence - driven sustainable development: Examining organizational, technical, and processing approaches to achieving global goals. *Sustainable Development*, 32(3), 2253–2267. <https://doi.org/10.1002/sd.2773>
- Kushwaha, A. K., Pharswan, R., Kumar, P., & Kar, A. K. (2022). How Do Users Feel When They Use Artificial Intelligence for Decision Making? A Framework for Assessing Users’ Perception. *Information Systems Frontiers*, 25(3), 1241–1260. <https://doi.org/10.1007/s10796-022-10293-2>
- Lawan, A. A., Muhammad, B. R., Tahir, A. M., Yarima, K. I., Zakari, A., Abdullahi II, A. H., ... & Lawan, S. (2023). Modified flipped learning as an approach to mitigate the adverse effects of

- generative artificial intelligence on education. *Education Journal*, 12(4), 136–143. <https://doi.org/10.11648/j.edu.20231204.14>
- Lee, A. V., Tan, S. C., & Teo, C. L. (2023a). Designs and practices using generative AI for sustainable student discourse and knowledge creation. *Smart Learning Environments*, 10(1), 1–18. <https://doi.org/10.1186/s40561-023-00279-1>
- Lee, S., Moon, W. K., Lee, J. G., & Sundar, S. S. (2023b). When the machine learns from users, is it helping or snooping? *Computers in Human Behavior*, 138, 107427. <https://doi.org/10.1016/j.chb.2022.107427>
- Lim, W. M., Gunasekara, A., Pallant, J. L., Pallant, J. I., & Pechenkina, E. (2023). Generative AI and the future of education: Ragnarök or reformation? A paradoxical perspective from management educators. *The International Journal of Management Education*, 21(2), 100790. <https://doi.org/10.1016/j.ijme.2023.100790>
- Liu, K., & Tao, D. (2022). The roles of trust, personalization, loss of privacy, and anthropomorphism in public acceptance of smart healthcare services. *Computers in Human Behavior*, 127, 107026. <https://doi.org/10.1016/j.chb.2021.107026>
- Liu, M., Ren, Y., Nyagoga, L. M., Stonier, F., Wu, Z., & Yu, L. (2023). Future of education in the era of generative artificial intelligence: Consensus among Chinese scholars on applications of ChatGPT in schools. *Future in Educational Research*, 1(1), 72–101. <https://doi.org/10.1002/fer.3.10>
- Liu, Q., & Khalil, M. (2023). Understanding privacy and data protection issues in learning analytics using a systematic review. *British Journal of Educational Technology*, 54(6), 1715–1747. <https://doi.org/10.1111/bjet.13388>
- Liu, S., Zhang, Y., Li, K., & Tan, J. (2019). Switching costs and benefits affect user resistance in project implementation: Roles of positive emotions and user traditionality. *Social Behavior and Personality: An International Journal*, 47(7), 1–10. <https://doi.org/10.2224/sbp.6908>
- Ma, X., & Huo, Y. (2023). Are users willing to embrace ChatGPT? Exploring the factors on the acceptance of chatbots from the perspective of AIDUA framework. *Technology in Society*, 75, 102362. <https://doi.org/10.1016/j.techsoc.2023.102362>
- Maheshwari, G. (2024). Factors influencing students' intention to adopt and use ChatGPT in higher education: A study in the Vietnamese context. *Education and Information Technologies*, 29(10), 12167–12195. <https://doi.org/10.1007/s10639-023-12333-z>
- Meireles, J. P. R., Azevedo, Y. G. P., Schwarz, L. A. D., & Gomes, H. B. (2019). Sunk cost effect in the decision-making process: An analysis with accounting and business administration students. *Revista Ambiente Contábil*, 11(2), 1–17. <https://doi.org/10.21680/2176-9036.2019v11n2ID18171>
- Memarian, B., & Doleck, T. (2023). Fairness, Accountability, Transparency, and Ethics (FATE) in Artificial Intelligence (AI) and higher education: A systematic review. *Computers and Education: Artificial Intelligence*, 5, 100152. <https://doi.org/10.1016/j.caeai.2023.100152>
- Monzer, C., Moeller, J., Helberger, N., & Eskens, S. (2020). User Perspectives on the News Personalisation Process: Agency, Trust and Utility as Building Blocks. *Digital Journalism*, 8(9), 1142–1162. <https://doi.org/10.1080/21670811.2020.1773291>
- Moorhouse, B. L., Yeo, M. A., & Wan, Y. (2023). Generative AI tools and assessment: Guidelines of the world's top-ranking universities. *Computers and Education Open*, 5, 100151. <https://doi.org/10.1016/j.caeo.2023.100151>
- Mutumukwe, C., Viberg, O., Oberg, L. M., & Cerratto-Pargman, T. (2022). Students' privacy concerns in learning analytics: Model development. *British Journal of Educational Technology*, 53(4), 932–951. <https://doi.org/10.1111/bjet.13234>
- Nißen, M., Selimi, D., Janssen, A., Cardona, D. R., Breitner, M. H., Kowatsch, T., & von Wangenheim, F. (2022). See you soon again, chatbot? A design taxonomy to characterize user-chatbot relationships with different time horizons. *Computers in Human Behavior*, 127, 107043. <https://doi.org/10.1016/j.chb.2021.107043>
- Niu, B., & Mvondo, G. N. (2024). I Am ChatGPT, the ultimate AI Chatbot! Investigating the determinants of users' loyalty and ethical usage concerns of ChatGPT. *Journal of Retailing and Consumer Services*, 76, 103562. <https://doi.org/10.1016/j.jretconser.2023.103562>
- Omrani, N., Rivieccio, G., Fiore, U., Schiavone, F., & Agreda, S. G. (2022). To trust or not to trust? An assessment of trust in AI-based systems: Concerns, ethics and contexts. *Technological Forecasting and Social Change*, 181, 121763. <https://doi.org/10.1016/j.techfore.2022.121763>
- Pereira, T., Limberger, D. P., & Ardigó, D. C. (2021). The moderating effect of the need for interaction with a service employee on purchase intention in chatbots. *Telematics and Informatics Reports*, 1, 100003. <https://doi.org/10.1016/j.teler.2022.100003>

- Pechenkina, E., & Branigan, E. (2022). Exploring staff and student experiences of learning management system transition. *Educational Research*, 65(1), 82–98. <https://doi.org/10.1080/00131881.2022.2147854>
- Petcharat, T., Jattamart, A., & Leelasantitham, A. (2023). A conceptual model to imply a negative innovation assessment framework on consumer behaviors through the electronic business platforms. *Journal of Retailing and Consumer Services*, 74, 103450. <https://doi.org/10.1016/j.jretconser.2023.103450>
- Popescu, A. R., & Schut, A. (2023). Generative AI in creative design processes: a dive into possible cognitive biases. *International Association of Societies of Design Research Congress 2023*, October, Politecnico di Milano, Italy,. <https://doi.org/10.21606/iasdr.2023.784>
- Priyambada, S. A., Usagawa, T., & Mahendrawathi, E. R. (2023). Two-layer ensemble prediction of students' performance using learning behavior and domain knowledge. *Computers and Education: Artificial Intelligence*, 5, 100149. <https://doi.org/10.1016/j.caeai.2023.100149>
- Rai, A. (2020). Explainable AI: From black box to glass box. *Journal of the Academy of Marketing Science*, 48, 137–141. <https://doi.org/10.1007/s11747-019-00710-5>
- Ratten, V., & Jones, P. (2023). Generative artificial intelligence (ChatGPT): Implications for management educators. *The International Journal of Management Education*, 21(3), 100857. <https://doi.org/10.1016/j.ijme.2023.100857>
- Sahlgren, O. (2023). The politics and reciprocal (re) configuration of accountability and fairness in data-driven education. *Learning, Media and Technology*, 48(1), 95–108. <https://doi.org/10.1080/17439884.2021.1986065>
- Santos, M. I. D. C., dos Santos, R. F., & Callado, A. A. C. (2021). The sunk cost effect on the decision-making process of students from higher education institutions. *Revista De Contabilidade e Controladoria*, 13(2), 113–132. <https://doi.org/10.5380/RCC.V13I2.79197>
- Samuelson, W., & Zeckhauser, R. (1988). Status quo bias in decision making. *Journal of Risk and Uncertainty*, 1, 7–59. <https://doi.org/10.1007/BF00055564>
- Shahzad, M., Xu, S., & Asif, M. (2024). Factors affecting generative artificial intelligence, such as ChatGPT, use in higher education: An application of technology acceptance model. *British Educational Research Journal*, 4084. <https://doi.org/10.1002/berj.4084>
- Shin, D. (2020). User Perceptions of Algorithmic Decisions in the Personalized AI System: Perceptual Evaluation of Fairness, Accountability, Transparency, and Explainability. *Journal of Broadcasting & Electronic Media*, 64(4), 541–565. <https://doi.org/10.1080/08838151.2020.1843357>
- Shin, D., Kee, K. F., & Shin, E. Y. (2022). Algorithm awareness: Why user awareness is critical for personal privacy in the adoption of algorithmic platforms? *International Journal of Information Management*, 65, 102494. <https://doi.org/10.1016/j.ijinfomgt.2022.102494>
- Shin, D., & Park, Y. J. (2019). Role of fairness, accountability, and transparency in algorithmic affordance. *Computers in Human Behavior*, 98, 277–284. <https://doi.org/10.1016/j.chb.2019.04.019>
- Slade, S., Prinsloo, P., & Khalil, M. (2019). Learning analytics at the intersections of student trust, disclosure and benefit. *Proceedings of the 9th International Conference on learning analytics & knowledge*, 235–244. <https://doi.org/10.1145/3303772.3303796>
- Steele, J. L. (2023). To GPT or not GPT? Empowering our students to learn with AI. *Computers and Education: Artificial Intelligence*, 5, 100160. <https://doi.org/10.1016/j.caeai.2023.100160>
- Sun, Z., Zang, G., Wang, Z., Zhao, H., & Liu, W. (2023). VCAs as partners or servants? The effects of information sensitivity and anthropomorphism roles on privacy concerns. *Technological Forecasting and Social Change*, 192, 122560. <https://doi.org/10.1016/j.techfore.2023.122560>
- Surahman, E., & Wang, T. H. (2022). Academic dishonesty and trustworthy assessment in online learning: A systematic literature review. *Journal of Computer Assisted Learning*, 38(6), 1535–1553. <https://doi.org/10.1111/jcal.12708>
- Sweeney, S. (2023). Who wrote this? Essay mills and assessment – Considerations regarding contract cheating and AI in higher education. *The International Journal of Management Education*, 21(2). <https://doi.org/10.1016/j.ijme.2023.100818>
- Tayan, O., Hassan, A., Khankan, K., & Askool, S. (2024). Considerations for adapting higher education technology courses for AI large language models: A critical review of the impact of ChatGPT. *Machine Learning with Applications*, 15, 100513. <https://doi.org/10.1016/j.mlwa.2023.100513>
- Taylor, S. H., & Brisini, K. S. (2024). Parenting the TikTok algorithm: An algorithm awareness as process approach to online risks and opportunities. *Computers in Human Behavior*, 150, 107975. <https://doi.org/10.1016/j.chb.2023.107975>

- Tekic, Z., & Füller, J. (2023). Managing innovation in the era of AI. *Technology in Society*, 73, 102254. <https://doi.org/10.1016/j.techsoc.2023.102254>
- Tsai, J. M., Cheng, M. J., Tsai, H. H., Hung, S. W., & Chen, Y. L. (2019). Acceptance and resistance of telehealth: The perspective of dual-factor concepts in technology adoption. *International Journal of Information Management*, 49, 34–44. <https://doi.org/10.1016/j.ijinfomgt.2019.03.003>
- Vecchiarini, M., & Somià, T. (2023). Redefining entrepreneurship education in the age of artificial intelligence: An explorative analysis. *The International Journal of Management Education*, 21(3). <https://doi.org/10.1016/j.ijme.2023.100879>
- Vimalakumar, M., Sharma, S. K., Singh, J. B., & Dwivedi, Y. K. (2021). ‘Okay google, what about my privacy?’: User’s privacy perceptions and acceptance of voice based digital assistants. *Computers in Human Behavior*, 120, 106763. <https://doi.org/10.1016/j.chb.2021.106763>
- Vlachogianni, P., & Tselios, N. J. (2022). Perceived usability evaluation of educational technology using the System Usability Scale (SUS): A systematic review. *Journal of Research on Technology in Education*, 54(3), 392–409. <https://doi.org/10.1080/15391523.2020.1867938>
- Vu, Adkins, M., & Henderson, S. (2019). Aware, but don’t really care: Student perspectives on privacy and data collection in online courses. *Journal of Open, Flexible and Distance Learning*, 23(2), 42–51. <https://doi.org/10.61468/jofdl.v23i2.350>
- Wang, Y. Y., Wang, Y. S., & Lin, T. C. (2018). Developing and validating a technology upgrade model. *International Journal of Information Management*, 38(1), 7–26. <https://doi.org/10.1016/j.ijinfomgt.2017.07.003>
- Watermeyer, R., Phipps, L., Lanclos, D., & Knight, C. (2024). Generative AI and the Automating of Academia. *Postdigital Science and Education*, 6(2), 446–466. <https://doi.org/10.1007/s42438-023-00440-6>
- Wu, I. L., Hsieh, P. J., & Wu, S. M. (2022). Developing effective e-learning environments through e-learning use mediating technology affordance and constructivist learning aspects for performance impacts: Moderator of learner involvement. *The Internet and Higher Education*, 55. <https://doi.org/10.1016/j.iheduc.2022.100871>
- Xin, X., Shu-Jiang, Y., Nan, P., ChenXu, D., & Dan, L. (2022). Review on A big data-based innovative knowledge teaching evaluation system in universities. *Journal of Innovation & Knowledge*, 7(3), 100197. <https://doi.org/10.1016/j.jik.2022.100197>
- Yilmaz, R., & Karaoğlu Yılmaz, F. G. (2023). The effect of generative artificial intelligence (AI)- based tool use on students’ computational thinking skills, programming self-efficacy and motivation. *Computers and Education: Artificial Intelligence*, 4, 100147. <https://doi.org/10.1016/j.caeai.2023.100147>
- Yuan, L. (2023). Where does AI-driven Education, in the Chinese Context and Beyond, go next? *International Journal of Artificial Intelligence in Education*, 34(1), 31–41. <https://doi.org/10.1007/s40593-023-00341-6>
- Zastudil, C., Rogalska, M., Kapp, C., Vaughn, J., & MacNeil, S. (2023). Generative AI in Computing Education: Perspectives of Students and Instructors. *2023 IEEE Frontiers in Education Conference (FIE)* (pp. 1–9). <https://doi.org/10.1109/FIE58773.2023.10343467>
- Zhang, C., Hu, M., Wu, W., Kamran, F., & Wang, X. (2025). Unpacking perceived risks and AI trust influences pre-service teachers’ AI acceptance: A structural equation modeling-based multi-group analysis. *Education and Information Technologies*, 30(2), 2645–2672. <https://doi.org/10.1007/s10639-024-12905-7>

Publisher’s Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.